UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

# Lecture 2A: Task Analysis & Design Methods

EIN 3241: Human Factors and Ergonomics
Spring 2026

January 21, 2026

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# Admin Issues

- Research study final reminder!
- Quiz 1
  - Monday January 26
  - Modules 1 and 2 (Including Friday's)
  - Homework 1 to practice
    - No submission needed!

# Recap

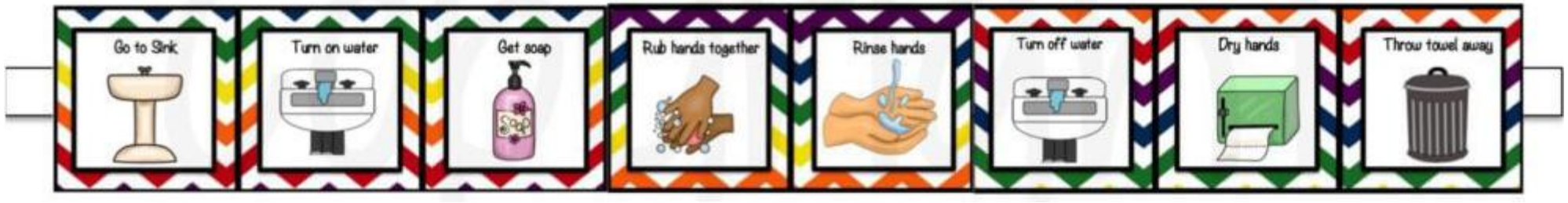
- The foundations of the science of ergonomics appear to go back as far as humans do…
- WWI&II: It was observed that complex military equipment could not be operated effectively or safely
- Post-War Expansion:
  - Birth of HF societies
  - Founding of OSHA & NIOSH
- Modern HFE:
  - Expansion into other domains
  - Department of Defense Military Standard
  - International Organization for Standardization
- Justifying an Ergonomic Program….

# By the end of this lecture, you will be able to:

- Define task analysis and explain its purpose
- Describe different types of task analysis methods
- Apply task analysis to a simple task
- Understand how task analysis informs system and interface design

# What is Task Analysis?

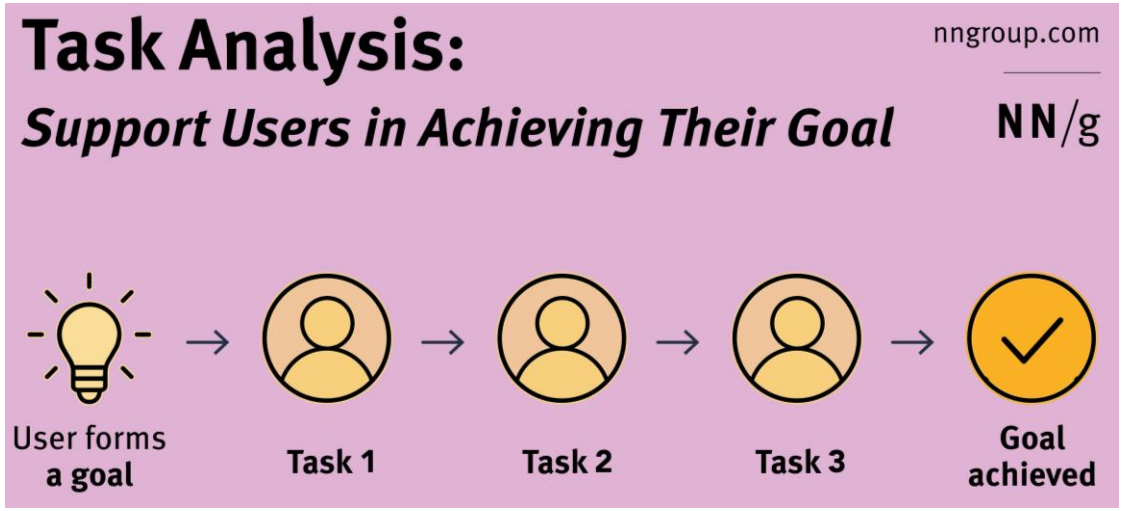
- A systematic method to study what users do to achieve goals
- Used to understand workflows, decisions, and errors
- Forms the basis for good design




**Basic example: Hand Washing Task Analysis**

# Why Task Analysis?

- Systems fail not just due to technical issues, but because tasks are misunderstood or poorly supported
- Task analysis helps:
  - Identify user goals and actions
  - Detect mismatches between design and human capability
  - Reduce errors and inefficiency



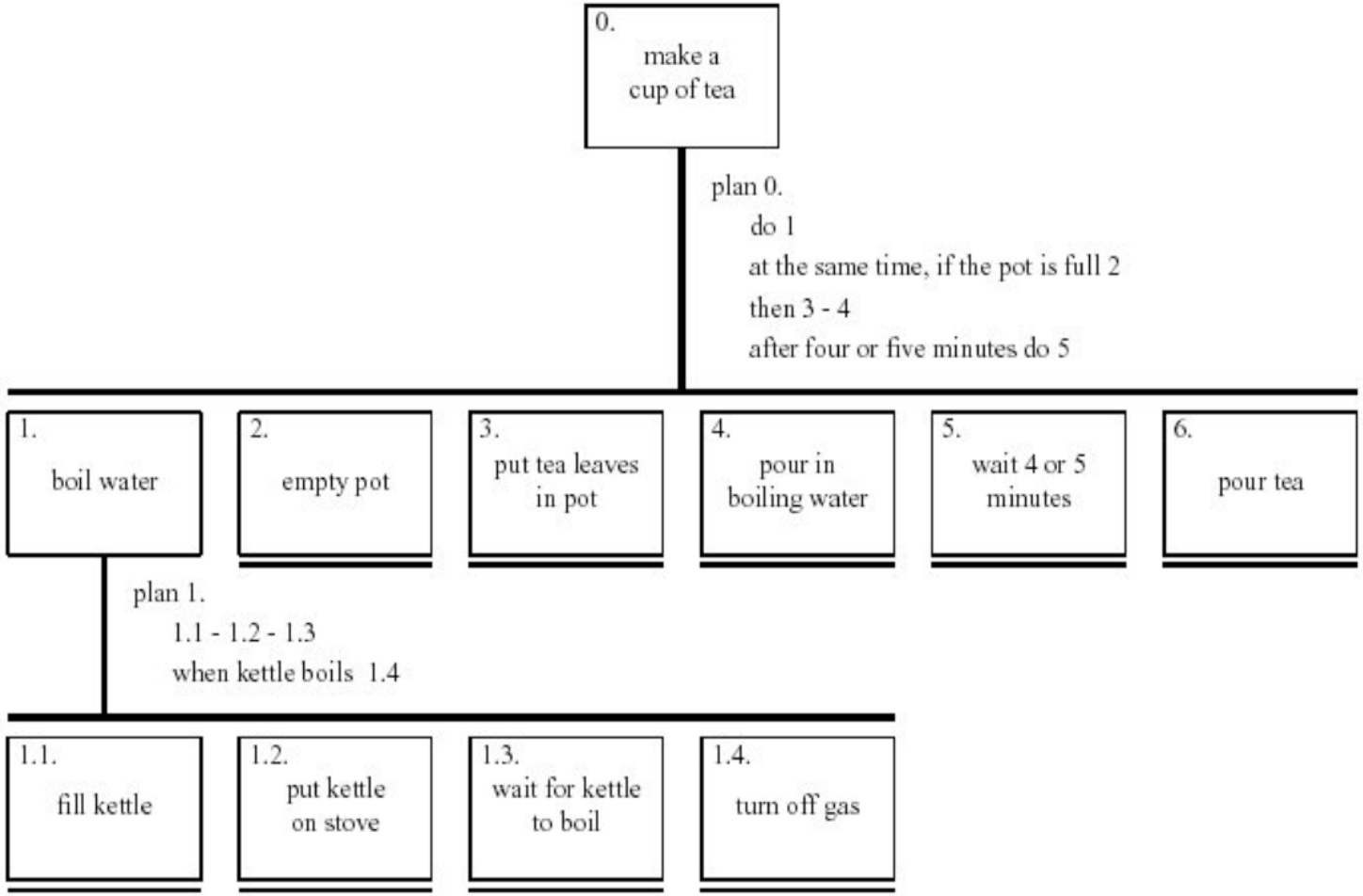
# Types of Task Analysis

- Hierarchical Task Analysis (HTA)
- Cognitive Task Analysis (CTA)
- Contextual Task Analysis
- Sequential Task Analysis (e.g., Flowcharts)
- Physical Task Analysis or Physical Demands Analysis
- Many more (Learning Analysis, Job/Performance Analysis…)

# Hierarchical Task Analysis (HTA)

- Plans are an essential element of describing tasks using HTA
- They describe conditions when subtasks are to be carried out to satisfy the tasks (or goals) being re-described
- You can account for considerable complexity in task by combining different plans
- Complex tasks can be better understood by representing them as a hierarchy of simpler plans
- Plans can describe various forms of fixed procedures, decision-making, cycles of activity, etc.  Plans also contain the timing and sequential relationships.

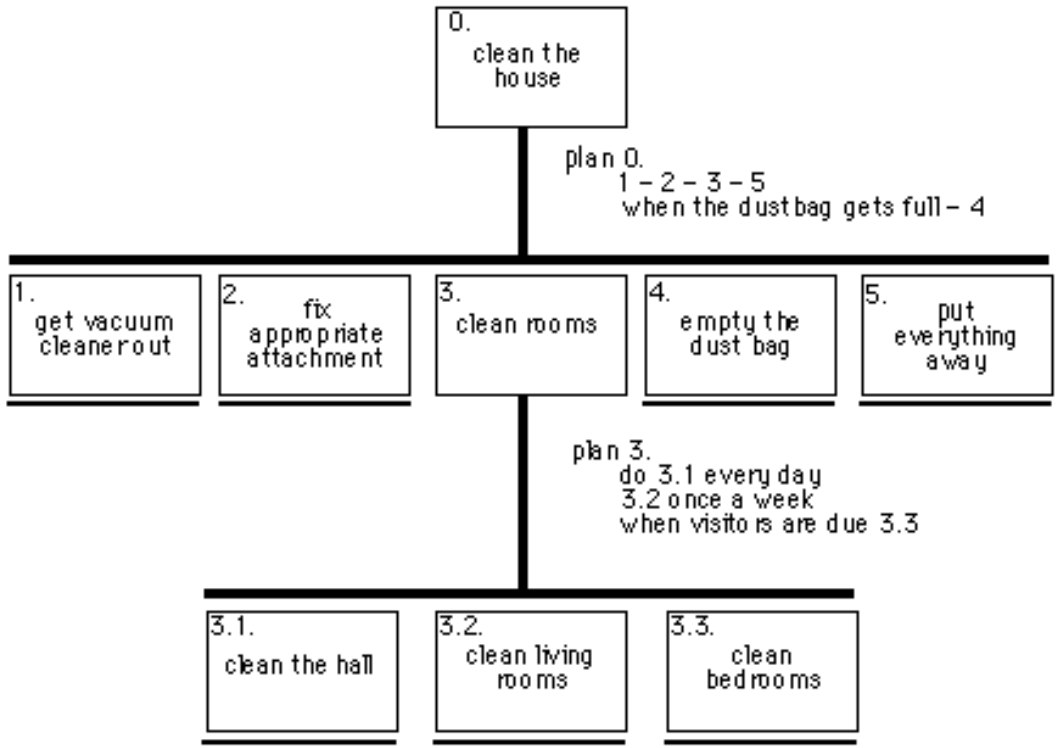
# Hierarchical Task Analysis (HTA)




Dix, A., Finlay, J., Abowd, G. and Beale, R., *Human-Computer Interaction*, Prentice Hall, 1993.

# Various Types of Plans

1. Sequential Plan
   - Tasks are performed in order
   - Plan: Do 1, then 2, then 3.
2. Concurrent Plan:
   - Tasks can be performed in parallel.
   - Plan: Do 1 and 2 at the same time if possible.
3. Optional Plan:
   - Some steps are conditional or user-preferred.
   - Plan: Do 1; then optionally do 2 if needed.
4. Contingent Plan:
   - Task flow depends on specific conditions or decisions
   - Plan: If A, do 1; if B, do 2.
5. Iterative Plan:
   - A step is repeated until a condition is met.
   - Plan: Repeat 1 until success.



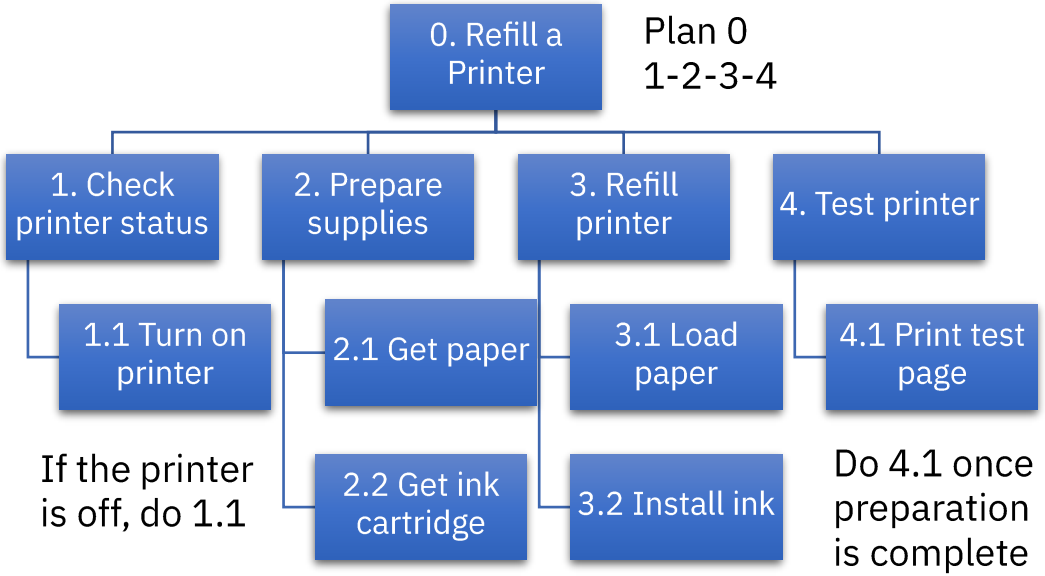
# HTA Limitations

- HTA plans cannot capture task coordination effectively if they are for an acquired skill (i.e. cognitive coordination).
- HTA describes observable behavior, but not internal thought processes like decision-making or attention.
- Difficult to specify tasks independent of current practice
  - devices
  - users
  - Contexts (environmental or situational influences)
- Plans can include contingencies, but HTA struggles with truly dynamic or adaptive task flows. In complex systems, the number of tasks becomes prohibitively large
- HTA is often designed for single-user analysis and may miss interactions between people and teams.

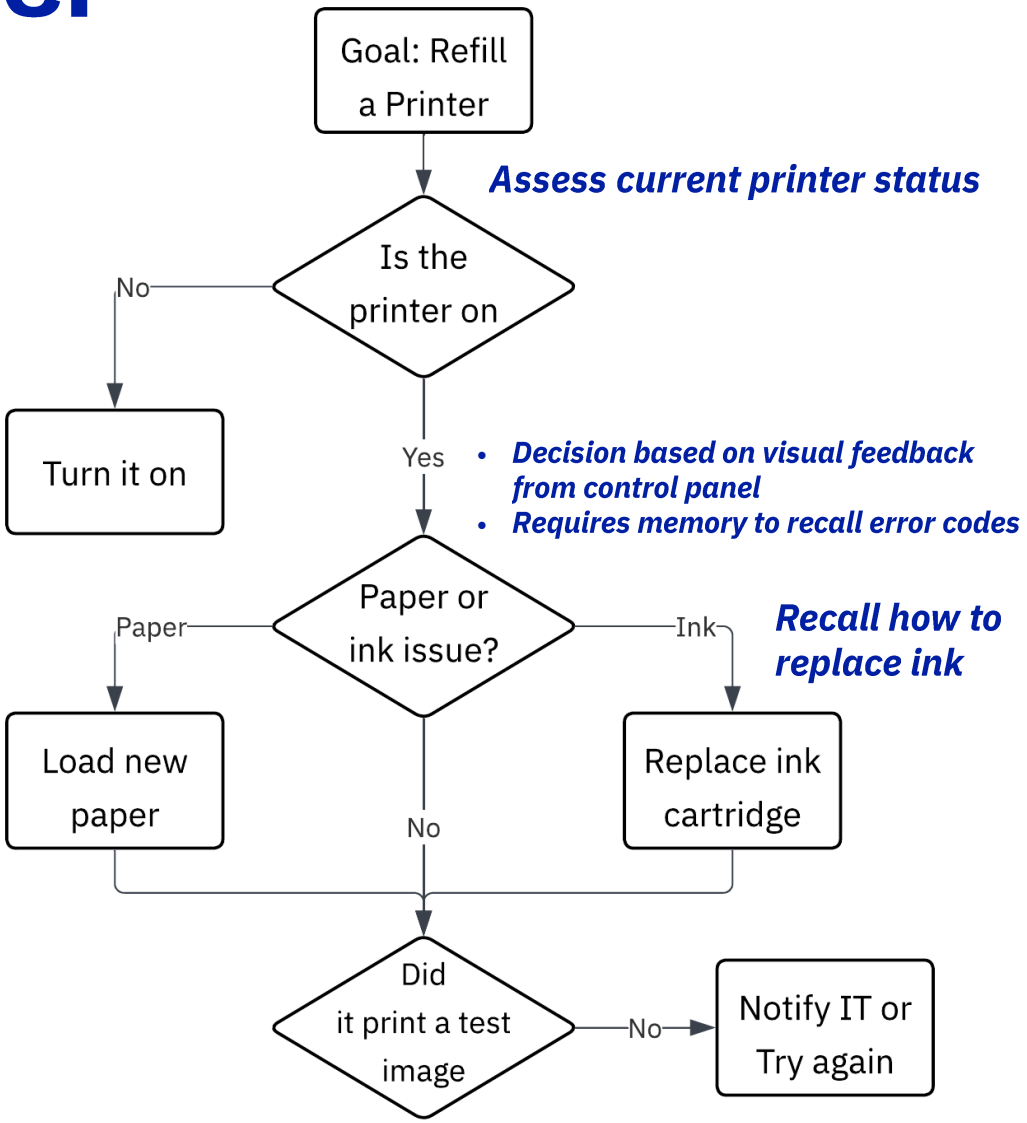
# Cognitive Task Analysis (CTA)

- Aimes at understanding tasks that require a lot of cognitive activity from the user
- Focuses on mental processes: attention, memory, decision-making
- Often used in safety-critical and high-stakes environments
- Methods: interviews, observations, think-aloud protocols…
- There are many ways to present task analysis, visually representing the user's mental process and awareness:
  - Graphs, flow charts, and sequence diagrams are appropriate options, including specific graphical references to crucial elements.
  - Whether color-coded or differently shaped information boxes, these elements allow us to easily spot human factors and their cognitive processes.
    - System actions
    - User actions
    - User decision making

# HTA vs CTA: Refilling a Printer



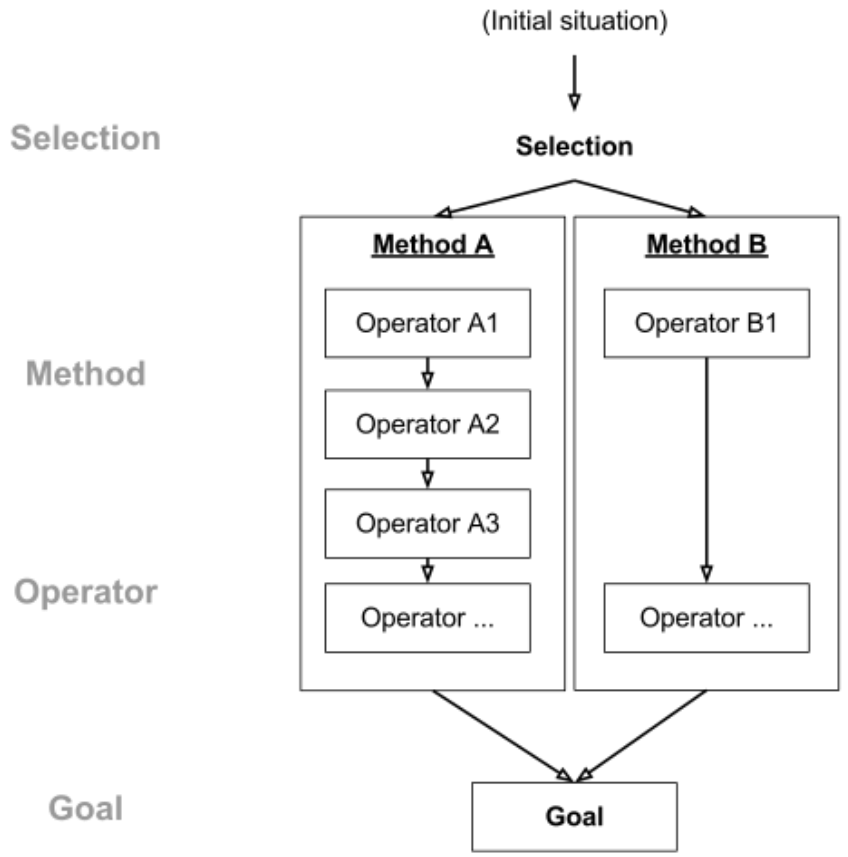

## HTA




## CTA

# GOMS (More in Appendix)

- GOMS is a model of human performance, and it can be used to improve human-computer interaction efficiency by eliminating useless or unnecessary interactions.
- GOMS → Goals, Operators, Methods, Selection
  - G: What the user wants to achieve
  - O: Basic perceptual, motor, or cognitive actions (e.g., press key, move mouse)
  - M: Procedures for accomplishing goals
  - S: How the user chooses between methods
- Can vary in granularity (degree to which O's are very detailed – hit key "a" vs. type your name)
- Good for modeling expert performance and predicting task time
- Assumes user is well-trained and tasks are routine
- Not great for modeling errors or novice behavior

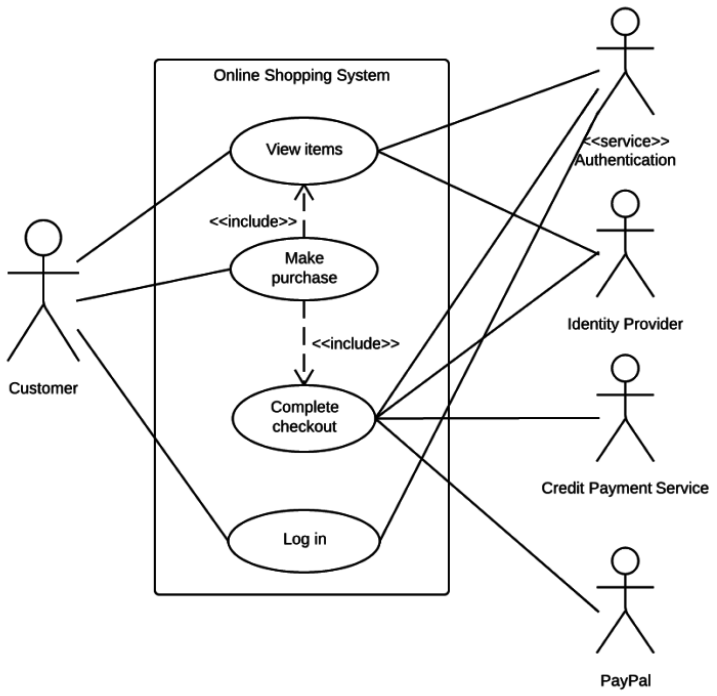


# How to Perform a Task Analysis

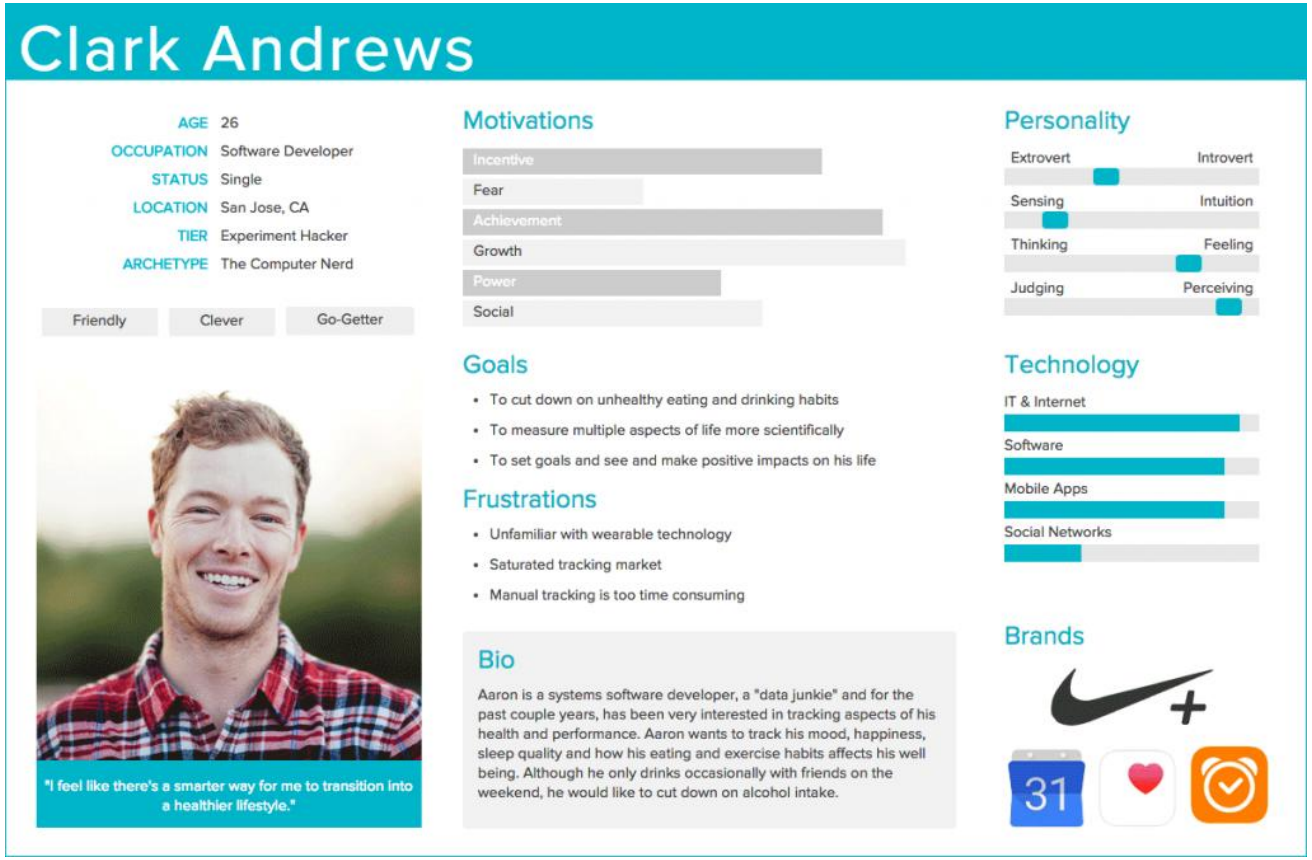
- Define the purpose and identify the data required
- Collect task data
- Interpret task data
- Innovate from task data:
  - **User identification and personas**
  - **Scenarios, user journeys, and use cases**
  - Environment and context analysis
  - Workload analysis
  - Safety and hazard analyses
  - Function allocation analysis

# Innovate from task data

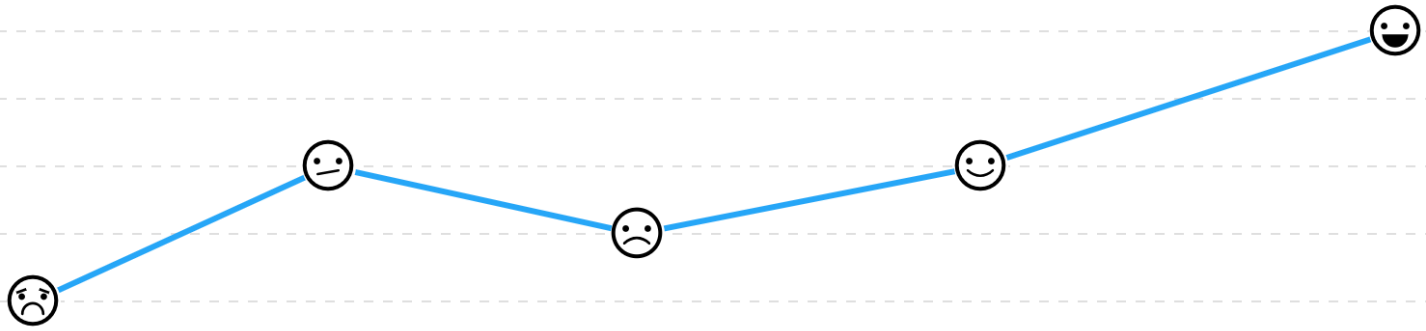
- Personas:
  - Capture a set of user characteristics (user profile)
  - Not real people, but synthesized from real users
  - Should not be idealized
  - Bring them to life with a name, characteristics, goals, personal background
- Scenarios
  - Concrete stories that concentrate on realistic and specific activities
  - Informal narrative, simple, ‘natural’, personal, non generalizable
- Use cases
  - Describes how a user uses a system to accomplish a particular goal
  - Assume interaction with a system
  - Assume detailed understanding of the interaction
- User journeys
  - Focus on the persona/user and their emotional states
  - A collection of scenarios or use cases that describe a user's interactions through the lifetime of interactions with the system



# Example Persona



| | TRIGGER | ARRIVAL | WARD | STAY IN WARD | DISCHARGE |
|---|---|---|---|---|---|
| Phase of Journey | Feels sick | Arrival in the hospital | Admitted as inpatient | Inpatient Stay | Discharge home |
| Actions | Calls daughter<br>Calls GP<br>Takes temperature, checks symptoms | Sees triage nurse<br>Waits for consult<br>Waits in ED and is admitted to bed in ED | Waits in ED<br>Wheeled up the ward<br>Introduced to NUM, orientated to ward | Talks to Dr<br>Receives treatment from allied health, dr, physio, nurses<br>Takes meds<br>Sleeps/eats | Care team organises services and medications for discharge<br>Organises discharge home with daughter<br>Travels home with daughter<br>Goes home |
| Thoughts | I don't feel well | Feels glad I can see someone<br>But feels confused if I need to stay in hospital<br>Who is my nurse?<br>Can someone tell my daughter? | I feel uncomfortable | I don't like the food here<br>I feel like I don't know which of the care team is coming to see me and when<br>My daughter doesn't know when to come<br>But I am feeling better | I feel glad to go home |
| Emotional Experience | | | | | |





https://uxplanet.org/personas-and-user-journeys-in-health-b4f4596f428d

# Innovate from task data

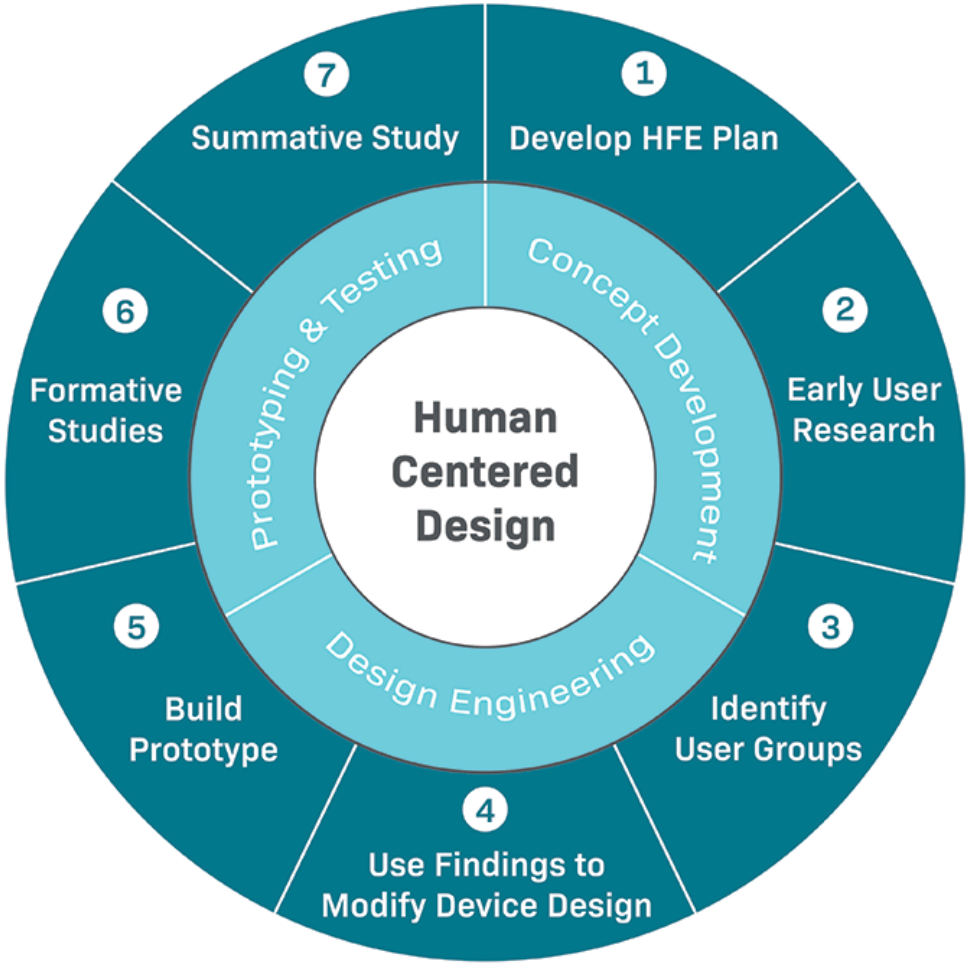
- User identification and personas
- Scenarios, user journeys, and use cases
- **Environment and context analysis**
- **Workload analysis**
- **Safety and hazard analyses**
- **Function allocation analysis**

# What did we learn today?

- Task Analysis: What and Why..
- Types of Task Analysis
  - Hierarchical Task Analysis (HTA)
  - Cognitive Task Analysis (CTA)
  - Contextual Task Analysis
  - Sequential Task Analysis
  - Physical Task Analysis
- Dived into HTA, CTA:
  - Plans…
  - GOMS…
- How to Perform a Task Analysis:
  - Define the purpose and identify the data required
  - Collect task data
  - Interpret task data
  - Innovate from task data
    - Personas
    - Scenarios
    - User Journeys
    - Use Cases

# Next Time…

- Evaluation in the Human Factors Design Cycle
- Differences between evaluation methods and how they support the human design cycle
- Understand the role of formative and summative human factors evaluations
- …



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LEADING THE CHARGE, CHARGING AHEAD

# Appendix Slides – Versions of GOMS

1. CMN-GOMS (Original Model):
   - "Model Human Processor" (Card, Moran, and Newell)
   - Requires detailed selection rules for method choice
2. KLM (Keystroke-Level Model):
   - A simplified version of GOMS
   - Predicts how long an expert user takes to perform a task
   - Only includes observable, low-level operators (button press, mouse click)
   - Useful for quick evaluations of interface efficiency
3. NGOMSL (Natural GOMS Language)
   - More formalized. Adds a structured, readable syntax (almost like pseudocode)
   - Rules of thumb for number of steps in a method
4. CPM-GOMS (Cognitive, Perceptual, and Motor GOMS)
   - Models parallel processing of cognitive, perceptual, and motor activities
   - Uses Critical Path Method (CPM) to predict task timing
   - Most realistic for expert users in high-speed, multitasking environments

# Appendix slides – GOMS

- Exploring the different GOMS versions:
- Task Example: Deleting a File Using a Mouse

1. CMN-GOMS:
   - Goal: Delete a file
   - Operators: Move mouse, click, read labels, etc.
   - Methods:
     - Method 1: Right-click → Select “Delete”
     - Method 2: Drag file to Trash
   - Selection Rule:
     - If user prefers menu navigation, use Method 1
     - If already pointing at file, use Method 21
   - Branching logic, multiple methods, rule-based choices

# Appendix slides – GOMS

2. KLM:
   - Predicts time by listing low-level physical actions
   - The keystroke-level model consists of six operators: the first four are physical motor operators followed by one mental operator and one system response operator
   - Only includes observable, low-level operators:
     - K = Keystroke or button press
     - P = Pointing with a mouse
     - H = Homing (hand movement between devices)
     - M = Mental preparation
     - R = System response time
   - In this example:
     - K (click file)
     - P (move to Trash)
     - K (drop)
     - M (mental prep)
     - R (system response)
   - Add fixed time estimates (e.g., K = 0.2s, P = 1.1s, etc.)
   - No goals/methods — just operator sequence + timing

# Appendix slides – GOMS

3. NGOMSL
   - Written in structured pseudocode:

```
Goal: Delete file
  Method: Using mouse
    Step 1: Locate file icon
    Step 2: Right-click on file
    Step 3: Select "Delete" from menu
    Step 4: Confirm deletion if prompted
```

   - Structured, readable; good for modeling novice use

# Appendix slides – GOMS

4. CPM-GOMS:
   - Adds parallel processing:
     - While the eye scans for the file (perception),the hand moves the mouse (motor), and the brain prepares to right-click (cognitive)
   - Could also use Gantt-like timeline to predict fastest path
   - Realistic for expert users doing tasks simultaneously

- Summary:

| Model | Captures Decisions? | Models Timing? | Models Learning? | Handles Multitasking? |
|---|---|---|---|---|
| CMN-GOMS | Yes | No | No | No |
| KLM | No | Yes | No | No |
| NGOMSL | Yes | Yes | Yes | No |
| CPM-GOMS | Yes | Yes | No | Yes |